Per-Track Momentum Re-calibration (κ Table) Tested with K^0S and Λ Mass Peaks

sPHENIX run3pp p+p 200 GeV, KFParticle V0s, DDSim templates

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287 segments, 32 runs, 284,648 events
κ(q, pT, η, φ) look-up table from runs 79509-79516 applied to every V0 daughter
Momentum scale and resolution refitted with and without κ
Tighter Λ selection for a cleaner Λ peak

Motivation and Questions

- run3pp K^0S peak ≈ 3 MeV/c^2 below PDG, width rises with pT much faster than in the full simulation
  - DDSim calibration absorbed it: s = 0.991, a = 3.75 %/(GeV/c), b = 1.40 % (current defaults)
- New input: per-track pT multiplier κ(q, pT, η, φ) derived on runs 79509-79516
  - pT → κ·pT for each daughter; η, φ unchanged; masses recomputed
- Questions
  - Does κ remove the scale offset (s → 1)?
  - How much of the fitted resolution was a charge-, η-, φ-dependent scale distortion?
  - Does a table from runs 79509-79516 also work for later runs?
  - Can a tighter Λ selection pin down the Λ peak (proton scale)?

Same candidates, three masses: KFP fit / recomputed without κ / recomputed with κ

Data Set and Processing

• DST_TRKR_TRACKS, ana581_2026p005_v001, release ana.581 (streaming, MVTX misalignment, CDB)
• One condor job per segment: KFParticle K^0S + Λ (no covariance cuts) → QA selection → mass vs pT
  – 480714: KFP / no κ / with κ histograms, 287/287 OK, KFP histograms bin-identical to the earlier production
  – 481293: + tight-Λ histograms + slim candidate trees (84 MB, cut studies now local), 287/287 OK, bin-identical
• Two periods: A = range of the κ table; B = all other runs (table applied as is)

period	runs	seg.	events	K^0S QA cand.	Λ + anti-Λ cand.
A	79509-79516 (8)	95	94,173	19,841	38,100
B	80904-81609 (24)	192	190,475	107,418	243,169

QA candidates = all masses in the histogram range (signal + background)

All validation checks pass: fill macro, reproducibility, κ table, grid edges

V0 Reconstruction and Selection

level	KFParticle (pre-cuts)	offline QA (data and DDSim templates)
event	PV from SvtxVertexMap, crossing-matched	$\|z_{PV}\| < 10$ cm
track	$p_T > 0.1$ GeV/c, ≥ 20 TPC hits, no χ^2/DCA cut	$\|\eta\| < 1.1$, $\|DCA_{xy}\| > 0.02$ cm (both daughters)
pair	daughter DCA < 0.5 cm; vertex χ^2, volume off	DCA12 < 0.2 cm (SV-constrained → no effect)
mother	DIRA > 0.8, mass window	DIRA > 0.995, $L_{xy} > 0.5$ cm
species	$K_S^0 \to \pi^+\pi^-$, $[\Lambda \to p\pi^-]cc$	Λ vs anti-Λ from the proton charge
tight Λ	—	+ DIRA > 0.9995, $\|DCA_{xy}(\pi)\| > 0.10$, $\|DCA_{xy}(p)\| > 0.03$ cm

- No covariance-, χ^2- or significance-based cut (DDSim cannot reproduce the covariance correlations)
- Cuts use the stored KFP values; they are not recomputed after κ
- All candidates kept (no best-candidate choice)

Fits use the QA histograms; the tight Λ selection is a separate study (slides 13-14)

The κ Look-up Table

- 72,800 nodes: q = ±1 × 112 pT (0.1-20 GeV/c) × 13 η × 25 φ
- pT_corr = κ · pT, κ = 1/(1 + ε + qδ)
  - ε: charge-even scale, qδ: charge-odd (sagitta-like) distortion
- Linear interpolation in (ln pT, η, φ), clamped at the edges
- Momentum vector scaled → η, φ unchanged; mass and V0 pT recomputed with PDG masses
- Mean κ applied (QA, period A): K^0S π^- 0.9979, π^+ 1.0163; Λ π 1.0077, p 1.0065

pT (GeV/c)	κ (q = −1)	κ (q = +1)
0.3	0.9952	1.0115
1.0	0.9725	1.0378
2.0	0.9379	1.0555

table nodes at η = 0, φ = 0

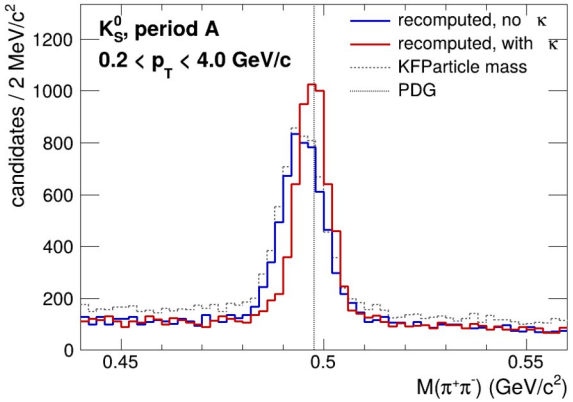
π^+ and π^- are shifted in opposite directions → the K^0S peak both moves and narrows

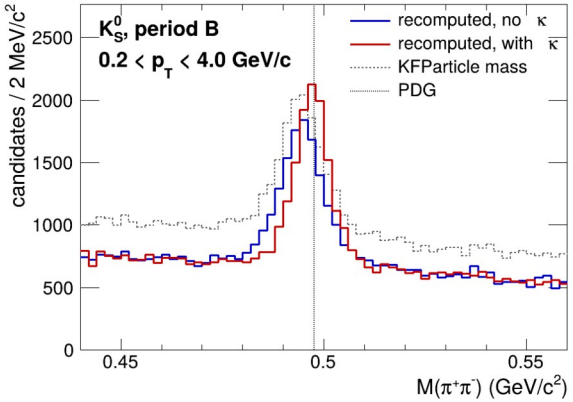
Momentum Scale / Resolution Fit

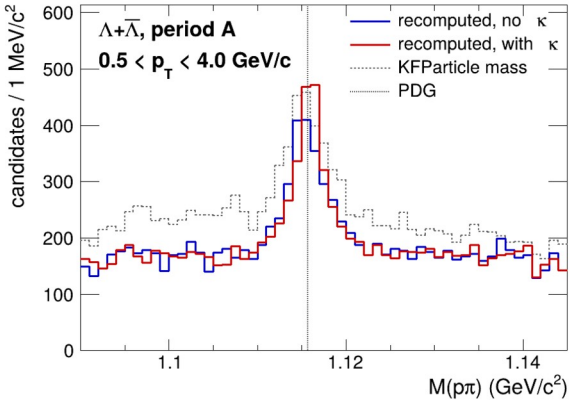
- Response per daughter: $p_{rec} = s \cdot p\,[1 + \delta_p(pT) + \sigma G]$, $\sigma/pT = \sqrt{((a\,pT)^2 + (b/\beta)^2)}$
  - s scale, a curvature term, b multiple scattering; δ_p = full-sim proton bias (protons only)
- Templates: DDSim, 6M decays per species, no momentum smearing, same QA cuts; fixed random numbers
- Data: K^0S in 7 V0-pT bins (0.2-4 GeV/c), Λ + anti-Λ in 5 bins (0.5-4 GeV/c, window 1.103-1.130 GeV/c^2)
- Per bin: N × template + quadratic background, Poisson likelihood, Baker-Cousins χ^2 summed over bins
- Grid: a 0-8 %/(GeV/c), b 0.4-4 %, s 0.975-1.015 (widened: κ moves s to ≈ 1)
- K^0S, Λ and joint fits; $\Delta\chi^2_\Lambda$ = Λ χ^2 at the K^0S best point − Λ minimum
- Periods A, B, all × KFP / no κ / with κ = 9 fits (+ 6 tight-Λ fits)

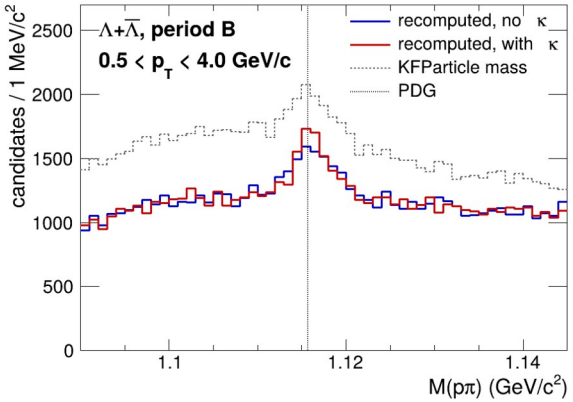
(a, b) are effective: they absorb every mass-width source DDSim does not model

Mass Distributions Before and After κ









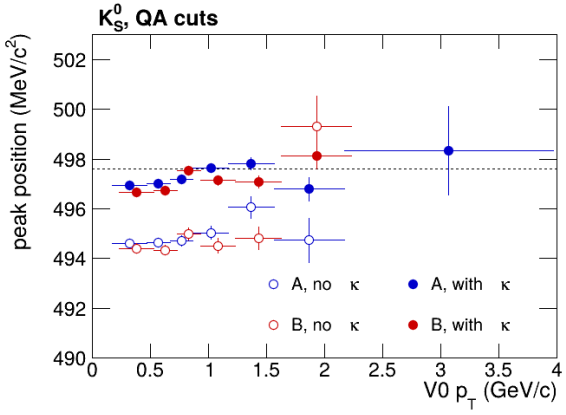
- K^0S mean (MeV/c²)
  - A 494.78 → 497.21
  - B 494.57 → 497.04
  - PDG 497.611
- K^0S σ (MeV/c²)
  - A 5.30 → 4.03
  - B 5.43 → 4.33
- Λ σ (MeV/c²)
  - A 2.16 → 1.63
  - B 2.69 → 1.88

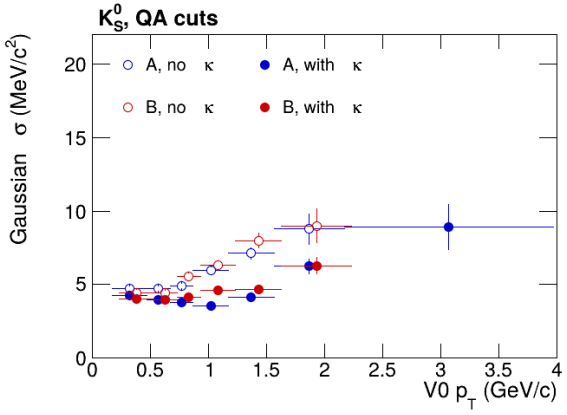
Gaussian fits, pT-integrated

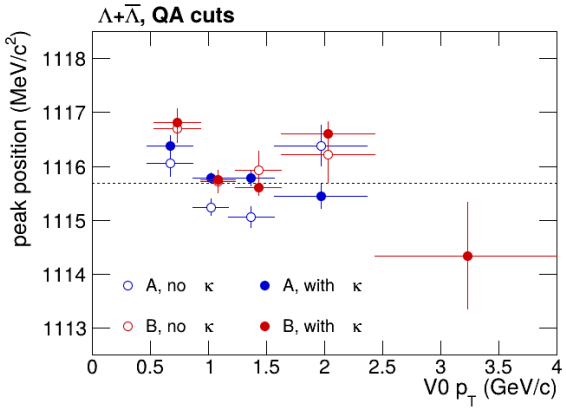
blue: no κ, red: with κ, grey dashed: KFP mass; dotted: PDG

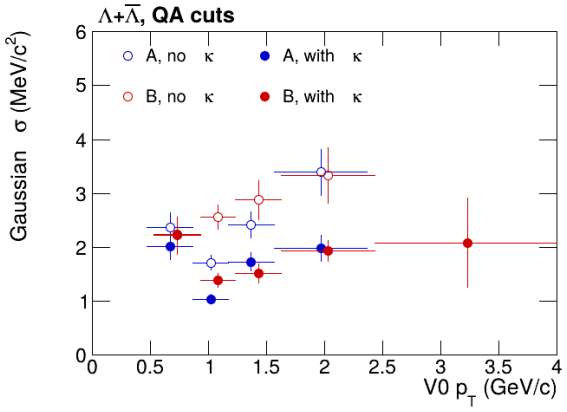
κ moves the K^0S peak to within 0.4-0.6 MeV/c² of PDG and narrows both peaks

Peak Position and Width vs pT









- With κ the K⁰S peak is within ≈ 1 MeV/c² of PDG in every pT bin (496.6-498.3)
- K⁰S width at V0 pT 1.6-2.2 GeV/c:
  - A 8.8 → 6.2 MeV/c²
  - B 9.0 → 6.3 MeV/c²
- Λ peak stays ≈ 0.2-0.4 MeV/c² above PDG
  - open issue: proton scale

open: no κ, full: with κ; blue A, red B; dashed PDG

Most of the high-pT width was a charge-dependent scale distortion

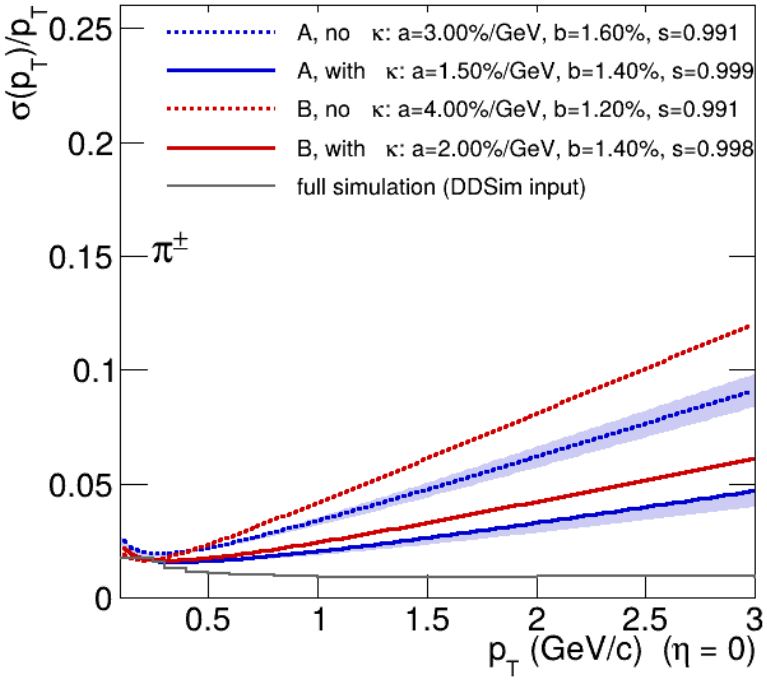
Fit Results: Scale and Resolution

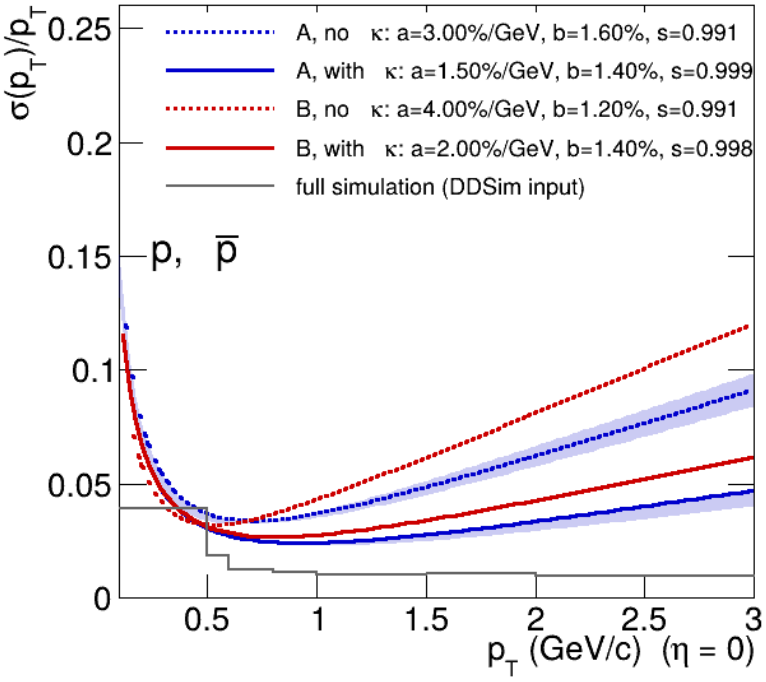
period	mass	a (%/GeV)	b (%)	s	χ^2_K/389	$\Delta\chi^2$_Λ
A	KFP	3.00	1.60	0.991	470.4	2.2
	no κ	3.00	1.60	0.991	464.5	3.4
	with κ	1.50	1.40	0.999	468.5	6.1
B	KFP	4.00	1.20	0.991	400.5	13.8
	no κ	4.00	1.20	0.991	473.1	30.5
	with κ	2.00	1.40	0.998	457.4	4.8
all	KFP	3.75	1.40	0.991	421.5	13.9
	no κ	3.50	1.40	0.991	462.1	25.0
	with κ	1.75	1.40	0.998	473.9	9.0

- Scale: s 0.991 → 0.999 (A), 0.998 (B); curvature term halved (0.50 A, 0.50 B)
- K^0S-Λ tension in B removed by κ: $\Delta\chi^2$_Λ 30.5 → 4.8

Period B keeps a larger curvature term after κ (2.00 vs 1.50 %/(GeV/c))

Single-Track Resolution



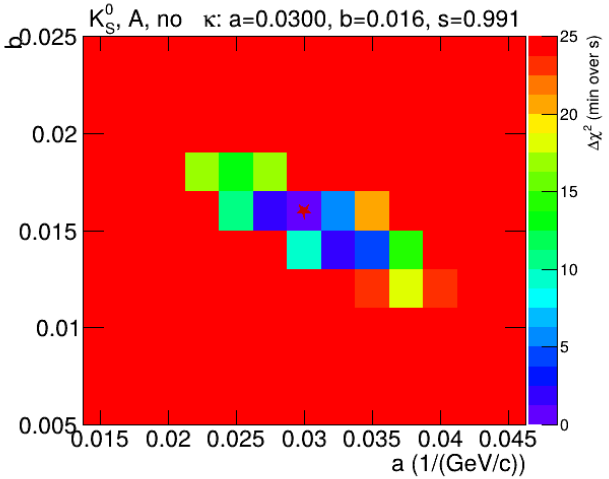


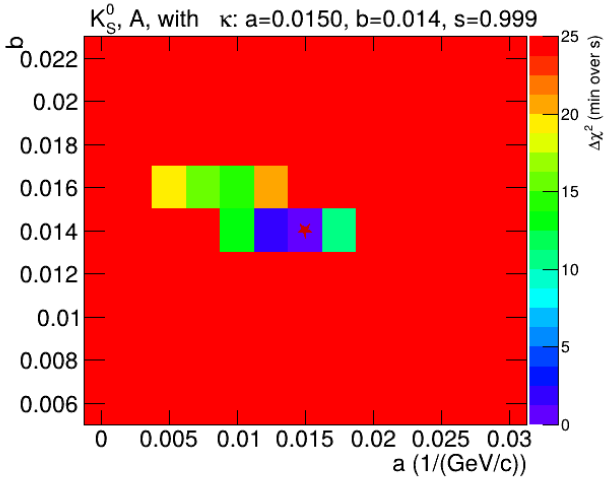
blue A, red B; dashed no κ, solid with κ; grey: full simulation

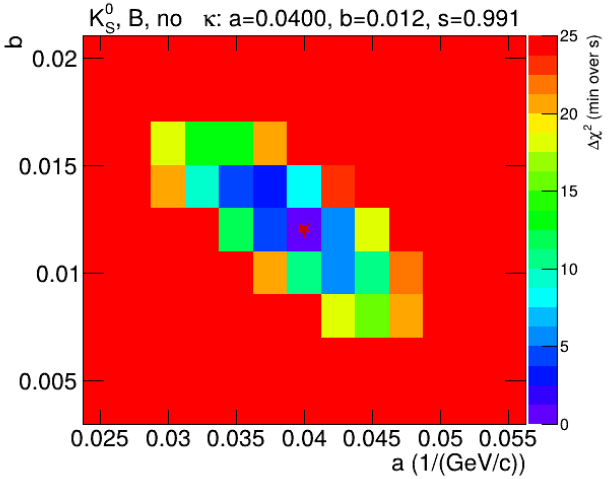
π σ/pT (%)	1 GeV/c	2 GeV/c	3 GeV/c
A: no κ → κ	3.4 → 2.1	6.2 → 3.3	9.1 → 4.7
B: no κ → κ	4.2 → 2.5	8.1 → 4.2	12.1 → 6.2

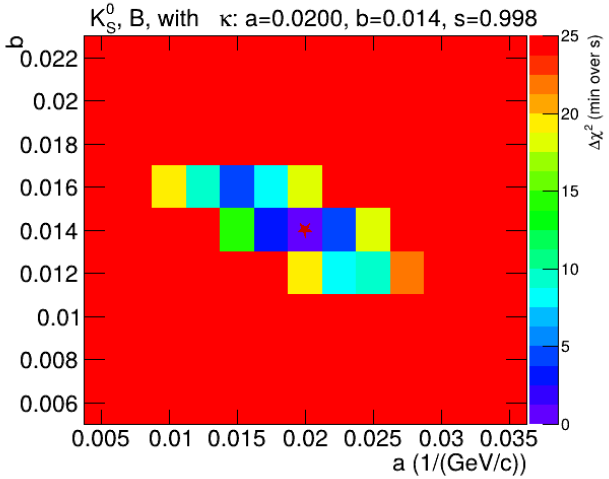
With κ the pion resolution at 2 GeV/c roughly halves; still above the full simulation

Δχ² Maps in (a, b)









- top: A, bottom: B
- left: no κ, right: with κ
- red star: best node
- a drops by 1.5 (A) and 2.0 (B) %/(GeV/c); b stays at 1.2-1.4 %
  - grid step (0.25 %/GeV in a) limits the interval resolution

Well-separated minima: the κ effect is far larger than the statistical uncertainty

Run Dependence

runs	a (%/(GeV/c)), KFP mass
79509 seg. 0-30	2.00
79509-79516 (A)	3.00
80904-80915	3.75
81556-81609	4.25

- K^0S QA candidates per event:
  - A 0.169-0.231
  - B 0.251-0.980
- a grows with the same ordering

- Hypothesis: luminosity-dependent distortion (e.g. TPC space charge) - not tested
- κ (from period A) removes about half of a in both periods; B remains higher
- Candidate next step: κ table derived on period-B runs, or per-run fits

Resolution is run dependent: quote DDSim parameters per period

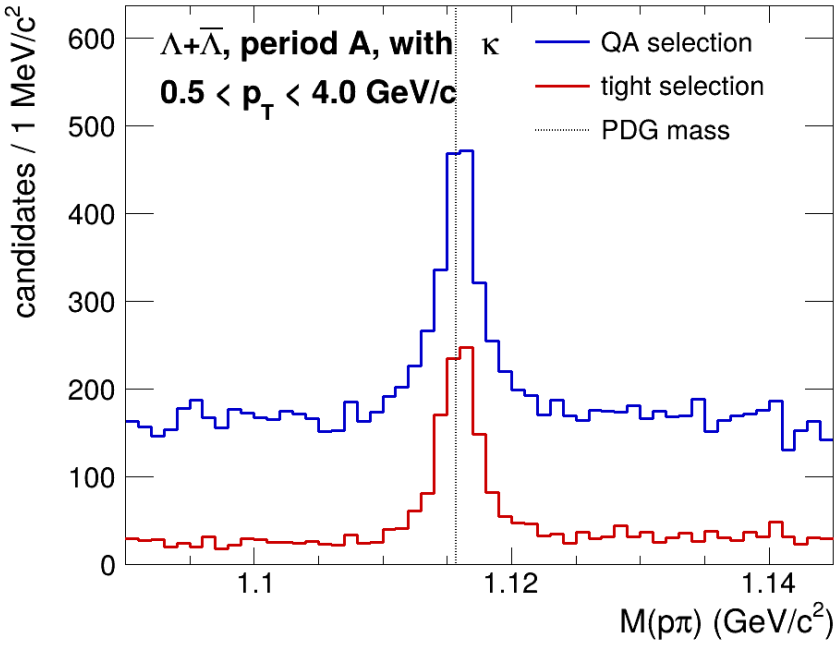
Tighter Λ Selection: Choice of Cuts

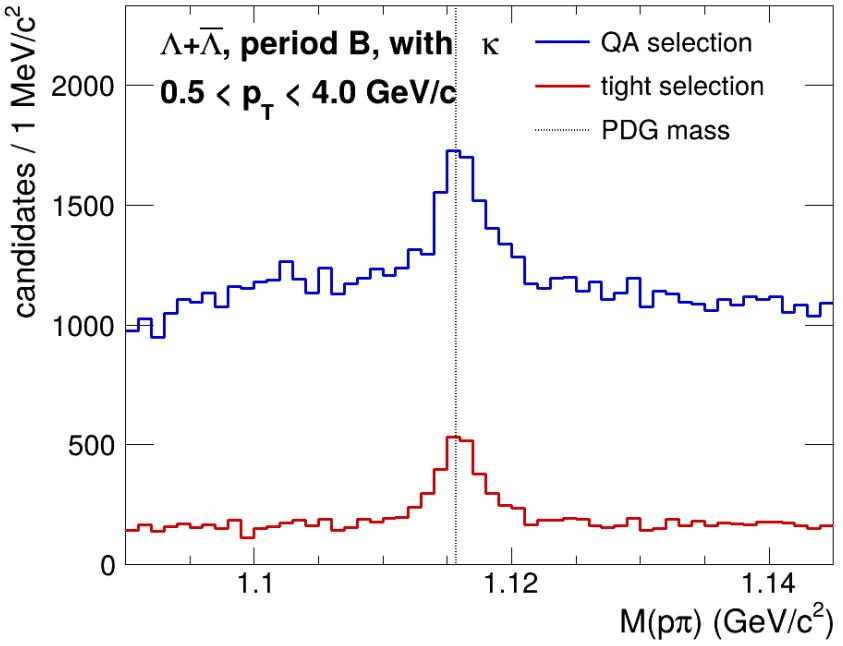
31 segments, KFP mass, pT 0.5-4	S	B	S/B	$S^2/(S+B)$	S kept
QA selection	363	1046	0.35	94	100 %
+ DIRA > 0.9995	332	383	0.86	154	91 %
+ L_xy > 2 cm (instead)	93	907	0.10	9	26 %
tight: DIRA > 0.9995, |DCA(π)| > 0.10, |DCA(p)| > 0.03	289	168	1.72	183	80 %

- 720 combinations of DIRA, L_xy, π and p |DCA_xy|; only cuts DDSim can model
- Bias test (DDSim selects before smearing): peak width ratio tight / QA
  - – K^0S: data 0.958 ± 0.042, DDSim 0.985; Λ: data 0.986 ± 0.149, DDSim 0.979
  - – DDSim Λ templates keep 0.76 of their candidates

Tight cut: $S^2/(S+B)$ × 1.96, no visible bias at the 5 % level (31 segments)

Tighter Λ Selection: Results





with κ	S/B	δ(peak) (MeV/c²)	S kept	s_Λ	a_Λ (%/GeV)	Δχ²_Λ
A: QA → tight	1.02 → 4.19	0.087 → 0.071	71 %	0.998 → 0.999	3.25 → 2.25	6.1 → 3.4
B: QA → tight	0.28 → 1.20	0.108 → 0.080	60 %	1.001 → 1.000	3.50 → 4.25	4.8 → 11.9
all: QA → tight	0.35 → 1.65	0.076 → 0.056	74 %	1.000 → 0.999	2.75 → 4.25	9.0 → 17.3

Used for the Λ scale (s_Λ = 0.999 A, 1.000 B), consistent with K^0S

Not for resolution: in B tight data widths exceed the tight templates (pointing modelling?)

Summary and Next Steps

- κ removes the momentum-scale offset: s 0.991 → 0.998; K^0S peak within 0.4-0.6 MeV/c^2 of PDG
- Curvature term halved: a 3.50 → 1.75 %/(GeV/c) (all), b ≈ 1.4 % unchanged
- Works outside its run range (period B) but B keeps a larger a (2.00 vs 1.50)
- K^0S/Λ consistent with κ; tight Λ selection gives s_Λ = 0.999-1.000 (± 0.001)
- Open
  - Λ peak 0.26 MeV/c^2 above PDG with κ: proton scale / δ_p double counting?
  - Tight Λ widths in period B (DIRA vs DCA cuts, local on the slim trees)
  - KFP vs recomputed mass: 20-25 % more background and different χ^2 in B

Decision: DDSim response = (a) uncorrected all-run defaults,

(b) κ-corrected per period (analyses apply κ), or (c) wait for a period-B table

Note: DataDrivenSim/kappa_note/build/main.pdf (28 pp)